

Changes in Opioid Use, Non-Prescription Drug Use and Help-Seeking Behaviours in Canada During the COVID-19 Pandemic

Michaela Bunakova¹, Amélie Quesnel-Vallée^{1,2}

¹Department of Sociology, McGill University, 885 Sherbrooke Street West, Montreal, Quebec, Canada, H3A 2T7

²Department of Epidemiology, Biostatistics and Occupational Health, McGill University, 1020 Pine Avenue West, Montreal, Quebec, Canada, H3A 1A2

Corresponding Author: Michaela Bunakova, MSc, Peterson Hall, McGill University, 3460 Mc Tavish St, Montreal, Quebec H3A 0E6. Email: michaela.bunakova@mail.mcgill.ca

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Availability of data and material: The CPSS6 PUMF dataset is publicly available for download from Statistics Canada. Data analysis and coding from R Studio, version 1.2.1335, are available upon request.

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Abstract

Objectives: COVID-19 has negatively impacted the lives and mental health and exacerbated barriers for people suffering from anxiety and substance abuse problems. This paper assesses patterns of opioid, non-prescription drug (NPD) use, and help-seeking behavior for substance misuse in Canada during the COVID-19 pandemic and explores how these changes relate to different sociodemographic characteristics.

Methods: We used data from the Canadian Perspectives Survey Series and binomial logistic regression models to estimate the associations between sociodemographic determinants of health and two primary outcomes: changes in drug use and help-seeking behavior for substance misuse.

Results: We found a significant association between increase in both opioid and NPD use and poor mental health outcomes during the COVID-19 pandemic. Compared to Canadians with 'excellent' mental health, those who self-rated their mental health as 'poor' were 8.1 and 5.7 times as likely to have increased their opioid and NPD use, respectively. The odds of unsuccessful help-seeking were over 19 and 14 times higher respectively for opioid and NPD users in 'poor', compared to 'excellent or good' mental health.

Conclusions: Our findings raise awareness of comorbid mental health and drug use problems among Canadians during the COVID-19 pandemic and point to the need for accessible and targeted interventions for those unable to access much-needed help during the pandemic.

Keywords: *Illicit Drugs, Opioids, Non-prescription Drugs, COVID-19, mental health, unmet care need*

1 Background

The novel coronavirus disease 2019 (COVID-19), has been spreading internationally and disrupting lives worldwide since it was declared a pandemic by the WHO in March 2020. Since then, the pandemic has spurred an ebullient research enterprise, not only in understanding and mapping the clinical and epidemiological evolution of the disease, but also regarding how social determinants of health, including poverty, smoke exposure or homelessness, affect COVID-19 outcomes (Dhar Chowdhury & Oommen, 2020; Abrams & Szefer, 2020). Surprisingly, however, research attention to the broader effects of the COVID-19 pandemic, including the effect of uncertainty, stress, and disruption to social services, on people's drug use patterns and help-seeking behavior has been limited. Most notably, Panchal and colleagues found that the pandemic ex-

acerbated mental health and substance abuse disorders, not only through its direct amplifying effect on those behaviours, but also because it imposed new barriers on the availability of public resources and access to needed help (Panchal & Kamal, 2021). Similarly, Horigian and colleagues document that COVID-19 particularly affected young people in the US, who now report significantly higher levels of comorbid loneliness, mental health problems and substance use (Horigian et al., 2021).

This paper has two objectives: (1) to estimate sociodemographic risk factors associated with increased consumption of opioids and non-prescription drugs during COVID-19, and (2) to document unmet needs for care and examine the extent to which drug-using Canadians seeking help for substance abuse or mental health problems experience difficulties in accessing public services during the pandemic.

It is especially pertinent to explore these two objectives in the

context of Canada, given its ongoing opioid crisis, escalating overdose deaths and polarizing public attitudes towards the issue (Korzinski, 2021). According to research from Angus Reid Institute, 70% of Canadians believe that opioid and non-prescription drug (NPD) addiction problems have worsened over the past year, while 48% fear the issue escalated in their own community (Korzinski, 2021). Since the onset of the COVID-19 pandemic, Canada witnessed increasing use of illicit injectable drugs and associated overdose deaths (Health Canada, 2020). Moreover, individuals with existing substance use disorders, opioid addictions and in recovery are even more vulnerable during the pandemic than before, because they may mistake SARS-COV-2 symptoms with addiction withdrawal symptoms and manage it by misusing opioids or other non-prescription drugs (Dunlop et al., 2020). However, personal characteristics and potential protective and risk factors associated with these changes in drug use in Canada are not currently understood. As its first objective, this paper aims to provide evidence of the magnitude and direction of the changes with reference to many established social determinants of health.

The Government of Canada took action to support people at increased risk of overdose and morbidities associated with unsafe injection drug use by expanding access to safer drug supply and providing the right for provinces to establish temporary Urgent Public Health Need Sites for safe drug use within, for example, supervised injection sites (Health Canada, 2020). However, the success of these new initiatives to support Canadians suffering from either licit or illicit drug abuse amidst the COVID-19 pandemic has not been evaluated from the perspective of drug users. To this end, the current paper pursues a second objective, an evaluation of the extent to which Canadians needed help with substance misuse or mental health but could not access or receive it.

Together, the two objectives aim to provide an overview of changes in the use of opioids and non-prescription drug as well as help-seeking behavior for substance misuse during the COVID-19 pandemic in Canada with an understanding of how individual sociodemographic characteristics affect both outcomes of interest. The paper hopes to contribute to existing evidence of the widespread effects of the COVID-19 pandemic on the Canadian society beyond the traditional focus on curbing the spread of COVID-19 or managing associated caseload and hospitalizations.

2 Methods

Data

We used data from a publicly available cross-sectional Public Use Microdata File (PUMF) dataset, 'Canadian Perspectives Survey Series 6: Substance Use and Stigma During the Pandemic' (CPSS6) administered by Statistics Canada. The CPSS6 frame for survey is Statistics Canada's pilot probability panel, created by randomly selecting a subset of respondents from the Labour Force Survey (LFS). The LFS sample is drawn from an area frame and is based on a stratified design using probability sampling (Statistics Canada, 2021a). Data for the CPSS6 dataset was collected using short online

surveys from January 25, 2021, to January 31, 2021. Survey participation was voluntary and the survey was designed to produce data representative of the Canadian population aged 15 and over from all provinces (Statistics Canada, 2021a). Overall, 7,242 CPSS panelists were eligible for the CPSS6 survey, of which 3,941 individuals responded, yielding a response rate of 54.4% (Statistics Canada, 2021a). A mixed mode approach was used for nonresponse follow-up, inclusive of mailed and e-mail reminders, where available (Statistics Canada, 2021a). All our analyses were weighted with LFS sample weights provided by Statistics Canada as part of the CPSS6 PUMF dataset.

Target population and analytic sample definition

Due to substantial missingness with respect to our main variables of interest (opioid and NPD use change during COVID-19) and a lack of complete overlap between the opioid and NPD user population, our paper includes two separate analytic samples. Change in cannabis use was excluded from this analysis because it no longer constitutes an illegal substance in Canada since the enactment of the 'Cannabis Act' in 2018 (Health Canada, 2018).

The target population is defined as Canadian residents aged 15 and older, who reported past use of opioids (analytic sample 1) and NPD (analytic sample 2). Based on the question 'In the past 30 days, have you used any prescribed or non-prescribed drugs containing opioids?' (analytic sample 1), and 'In the past 30 days, did you use any non-prescribed drugs?' (analytic sample 2), individuals who responded 'Yes' or 'No, not during the past 30 days' were included in our analytic samples. Individuals who responded 'Never used any opioids' (analytic sample 1) and 'Never used any non-prescribed drugs' (analytic sample 2) were excluded from analysis, because their responses were considered as 'valid skip' of our main outcome variables of interest. We do not make sample exclusion based on age for two reasons. Firstly, illicit drug possession, cultivation and use, exclusive of cannabis, is considered a criminal offence for all age groups across Canada, as legislated by The Controlled Drugs and Substances Act (Government of Canada, 1996). Secondly, drugs including opioids can be prescribed irrespective of age.

Only complete cases with regards to all main variables were used in the regressions. Missing values were uniformly re-coded and excluded from the statistical analysis using list-wise deletion, which significantly affected the sample size of both our two separate analytic samples. This is because only 9% and 7% of CPSS6 respondents reported previous opioid (N=346) and NPD use (N=308), respectively, and were therefore deemed eligible for inclusion in our study. It is important to note that in the case of Analytic Sample 1, the outcome of interest, change in opioid use, covers the use of both prescribed and non-prescribed opioids, and therefore ostensibly includes both legal and illicit opioid consumption. In the case of Analytic Sample 2, NPD use outcome is solely interested in misuse of non-prescribed substances, such as cocaine or hallucinogens.

Informative valid skip and treatment of missing values: While

in the main analysis, we considered individuals, who ‘skipped’ the questions related to drug use change or designated that these questions were ‘not applicable to them’, as ‘missing’, and excluded them from the analysis, we also hypothesize that this ‘missingness’ is informative.

Given this pattern of missingness, CPSS6 respondents excluded from our analytic samples 1 and 2 may either be those who never used opioids and NPDs, and hence questions about consumption changes during COVID-19 were not relevant to them (valid skip), or they may not wish to share their consumption patterns. Therefore, we could not consider missingness as random throughout our analytic samples. To exploit the possibility of valid skip to tell an additional story about potential protective factors associated with lower risk of opioid or NPD use, we chose to conduct two additional binary regression analyses using two new analytic samples, whereby we recoded both opioid and NPD use change variables into binary variables for valid responses (i.e. individuals who reported a history of past opioid or NPD use in a previous question) and valid skips (i.e., individuals who reported no drug use in a previous question). Therefore, in this additional analysis, we included individuals previously considered missing. This allowed us to construct two additional analytic samples of 3,932 and 3,934 individuals each, to test for characteristics associated with any level of opioid or NPD use, compared to a proxy for no use of opioids or NPDs.

Variables

Three dependent variables were included in the analysis, two of which were used towards Objective (1) to assess changes in opioid (analytic sample 1) and NPD use (analytic sample 2) during the COVID-19 pandemic among individuals who reported past opioid and NPD use.

One dependent variable was used towards Objective (2) to explore inability to find help for substance use, emotions, or mental health (included in both analytic samples as a proxy for unmet needs). This dependent variable was explored separately for both analytic samples, due to the possibility that patterns of unmet needs may differ by substance use type (i.e., among opioid and NPD users).

See Appendix A for detailed descriptions of the CPSS6 questionnaire for the dependent variables.

Independent variables consisted of eight sociodemographic respondent characteristics, namely age, sex, immigration status, marital status, highest education level, employment status, rural/urban indicator and mental health status. These covariates were chosen with reference to established research evidence on determinants of health behavior and drug use. Poor mental health is one of the most common comorbidities and a strong predictor of increased risk for both illicit drug initiation and drug use frequency (Burns et al. 2004, Burns and Teesson 2002, Stambaugh et al. 2017).

Similarly, research suggests that gender and sex also play a role in observed patterns of drug abuse and dependence, with men experiencing higher risk of hard drug initiation and rates of substance use overall, compared to women, while women

appear to be more susceptible to dependence and addiction to both hard and soft drugs (Cotto et al. 2010, Moon et al. 1999). Recent research also revealed that immigration status, education level, marital status, age, employment status and place of residence may also influence health behaviors and illicit drug use (Peña et al. 2008, McDonald and Kennedy 2004, Haider et al. 2009, DeSimone 2002, Cronk and Sarvela 1997, Scott et al. 2010, Musher-Eizenman, Holub, and Arnett 2003, Vang et al. 2017).

A list of descriptive statistics of all variables of interest for the full CPSS6 dataset as well as our two analytic samples is presented in Table I below.

Independent variables		N° (Full CPSS6)	N° (Analytic Sample 1)	N° (Analytic Sample 2)
Opioid use change	Same/Decreased (reference)	318 (8)	318 (92)	/
	Increased	28 (1)	28 (8)	/
	Not applicable/ Valid skip	3595 (91)	0 (0)	/
NPD use change	Same/Decreased (reference)	269 (7)	/	269 (87)
	Increased	39 (1)	/	39 (13)
	Not Applicable/Valid Skip	3633 (93)	/	0 (0)
Unsuccessful help-seeking	No (reference)	3465 (88)	275 (80)	242 (79)
	Yes	450 (12)	68 (20)	66 (21)
Mental health	Excellent/good (reference)	3061 (77)	230 (66)	197 (65)
	Fair	678 (18)	73 (21)	71 (23)
	Poor	195 (5)	43 (13)	38 (12)
Immigration status	Born in Canada (reference)	3260 (83)	299 (86)	256 (83)
	Landed Immigrant	681 (17)	47 (14)	52 (17)
Education level	College or lower (reference)	2219 (56)	202 (58)	161 (52)
	University education	1722 (44)	138 (42)	147 (48)
Place of residence	Rural (reference)	825 (21)	60 (17)	60 (19)
	Urban	3116 (79)	285 (83)	248 (81)
Employment status	Employed (reference)	2295 (60)	180 (52)	184 (62)
	Unemployed	1545 (40)	153 (48)	113 (38)
Sex of respondent	Male (reference)	1808 (46)	142 (41)	146 (48)
	Female	2133 (54)	204 (59)	162 (52)
Marital status	Married/Common-Law (reference)	2477 (63)	194 (56)	169 (51)
	Single / Widowed / Divorced	1464 (37)	152 (44)	145 (49)
Age	15-34 (reference)	638 (16)	53 (15)	69 (22)
	35-64	2200 (56)	196 (57)	167 (55)
	65+	1103 (28)	97 (28)	72 (23)
Observations:		3,941	346	308

Figure 1. Descriptive Statistics for CPSS6, Analytic Sample 1 and Analytic Sample 2 (category-specific sample proportions in parentheses).

Analytic sample 1: Opioid use change during COVID-19. The distribution of key analytic sample characteristics revealed the following information about our independent variables: 86% of respondents were born in Canada, 42% hold bachelor's degree or higher, 83% are urban dwellers, 48% are unemployed, 59% are female, 56% are partnered (married or common law). Of the sample population, 66% consider themselves in excellent or good mental health while 13% assess their mental health as poor. Overall, young adults between 15 and 34 years old represent 15% of the sample, 57% are between 35 and 64 years, and the 65+ age group is represented by 28% of the sample.

Moreover, Table I reveals that there is a difference between the full CPSS6 sample and analytic sample 1 of substantial magnitude with respect to unsuccessful help-seeking for emotions, mental health and substance misuse, self-reported mental health, employment status and marital status. Analytic sam-

ple 1 respondents appear to be more likely to seek help, but be unable to find it, compared to the full CPSS6 dataset. They are also more likely to be in poor mental health, unemployed and single, widowed, or divorced. By design of inclusion criteria, analytic sample 1 respondents are all present or past opioid users, and therefore the difference in descriptive statistics with respect to opioid use was expected.

Analytic sample 2: NPD use change during COVID-19. From Table I, we see that 83% of respondents were born in Canada, 48% hold a university degree, 81% live in urban areas, 38% are unemployed, 52% are female, 51% are partnered (married or common law). Among the analytic sample 2 respondents, 65% consider themselves in excellent mental health while 12% assess their mental health as poor. Young adults between the age of 15 and 34 represent 22% of the sample, 55% are between 35 and 64 years, and the 65+ age group is represented by 23% of the sample. We can see that compared to the full sample, analytic sample 2 respondents significantly differ with respect to NPD use change, unsuccessful help-seeking for substance misuse, mental health, marital status, and age. Overall, the respondents are younger, more likely to be single, widowed or divorced, in poor mental health and unable to find help for substance misuse. By design of inclusion criteria, analytic sample 2 respondents are all present or past NPD users, and therefore the difference in descriptive statistics with respect to opioid use was expected.

Analysis

Drug use change and unsuccessful help-seeking

We used a set of four binary logistic regressions within our two separate analytic samples to address our two primary objectives using data from CPSS6 and sample weights. The logistic regressions were estimated as a function of the eight sociodemographic characteristics presented in Table I. For NPD and opioid use outcomes, we assessed whether consumption was likely to ‘increase’ as opposed to ‘decrease or stay the same’. For help-seeking, the binary outcome was a ‘yes’ versus ‘no’, responding to whether the respondents were unsuccessful in finding help for substance misuse or mental health problems. The reference category was ‘no’. Regression results are presented graphically using odds ratio forest plots with 95% confidence intervals. A tabular form of the results can be found in Appendix B. Additional separate models for demographic and socioeconomic characteristics for each outcome are included in Supplemental Material (see Appendix C). All data analysis was conducted using R and R Studio, version 1.2.1335.

NPD and opioid use change response/valid skip analysis

We conducted an additional binomial logit regression analysis to check for possible systemic differences with regards to our predictor variables between the sample of respondents who reported past drug use and those who previously stated that they have not used drugs in the past (valid skip on our outcome variables of interest). This analysis offers a proxy

analysis for determinants of ‘no drug use’ versus ‘any level of drug use’. This is because we expect individuals who have never used opioids or NPD drugs to either skip the question or report that it is not applicable to them, thereby indicating that they used no opioids or NPDs, both before and during the pandemic.

The results from this proxy analysis, however, need to be reported with caution, for two alternative scenarios could have happened; firstly, individuals who use drugs but are not comfortable with sharing this information may have skipped the questions, and secondly, individuals who do not use drugs may have misinterpreted the framing of the question and report that their drug use ‘stayed the same’, that is, at zero consumption (see Table II below for descriptive statistics related to these additional samples). From the tables, we can see that there are significant differences between respondents and non-respondents of both opioid and NPD questions with respect to self-reported mental health, marital status, and age.

From the descriptive statistics, we can see that those who reported prior opioid and NPD use differ from those who did not in terms of their poorer mental health and higher probability of being single, divorced, or widowed, respectively. Moreover, respondents who reported prior opioid use were more likely to be unemployed, while those who reported prior NPD use were younger than those who did not.

Independent variables		Opioid use change		NPD use change	
		Valid skip (%)	Valid survey response (%)	Valid skip (%)	Valid survey response (%)
Mental Health	Excellent/Good	2830 (79)	230 (66)	2862 (79)	197 (64)
	Fair	604 (17)	73 (21)	607 (17)	71 (23.2)
	Poor	152 (4)	43 (13)	157 (4)	38 (12.4)
Immigration status	Born in Canada	2959 (83)	299 (86)	3002 (83)	256 (83)
	Landed immigrant	634 (17)	47 (14)	629 (17)	52 (17)
Sex	Male	1665 (46)	142 (41)	1662 (46)	146 (47)
	Female	1928 (54)	204 (59)	1969 (54)	162 (53)
Education level	College or lower	2010 (56)	208 (60)	2058 (57)	161 (52)
	University education	1583 (44)	138 (40)	1573 (43)	147 (48)
Place of residence	Rural	764 (21)	61 (18)	764 (21)	60 (20)
	Urban	2829 (79)	285 (82)	2867 (79)	248 (80)
Employment status	Employed	2115 (59)	180 (52)	2111 (60)	184 (62)
	Unemployed	1390 (41)	153 (48)	1430 (40)	113 (38)
Marital status	Married/Common Law	2283 (64)	194 (56)	2307 (63)	169 (55)
	Single/Widowed/Divorced	1310 (36)	152 (44)	1324 (37)	139 (45)
Age group	15-34	585 (16)	53 (15)	569 (16)	69 (22)
	35-64	2004 (55)	196 (56)	2032 (56)	167 (54)
	65+	1004 (29)	97 (29)	1030 (28)	72 (23)
Total Observations (3,932)		3,586	346	3,626	308

Figure 2. Descriptive Statistics for Opioid and NPD Response/Valid Skip analytic samples (category-specific sample proportions in parentheses).

3 Results

Changes in opioid and NPD use pre- and during COVID-19

In Figure 1 below, we see that over 83% of individuals who used opioids before the pandemic did not change their consumption patterns, while roughly 8% of opioid users both increased, and decreased their consumption compared to their pre-pandemic consumption levels. We also see that 79% of in-

dividuals who used NPD before the pandemic did not change their consumption patterns, while almost 13% of NPD users increased, and 8% decreased their consumption compared to the pre-pandemic levels. Overall, while roughly 8% of both opioid and NPD users in Canada decreased their consumption during the pandemic, the risk of increased consumption was significantly higher for NPD users than opioid users.

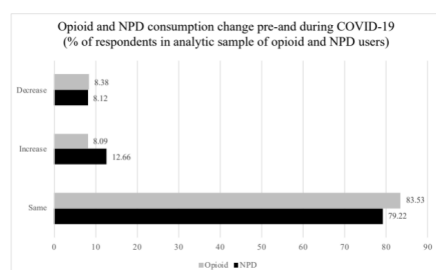


Figure 3. Bar plot of opioid and NPD use change pre and during COVID-19.

Opioid use increase during COVID-19 and unmet treatment needs for substance misuse and mental health among opioid users

Figures 2 and 3 below present the associations between opioid use change, ability to access treatment among opioid users and eight predictor variables (binary regression results tables are available in Appendix B).

Odds of opioid use increase. Poor mental health was the only predictor of increased opioid use during the COVID-19 pandemic. Compared to Canadians who self-rated their mental health as excellent or good, those with ‘poor’ mental health are expected to be 8 times more likely to have increased their opioid use relative to decreasing it or keeping it constant at pre-pandemic levels (8.11, 95% CI 2.99–22.75). We found no effect of immigration, marital and employment status on the odds of increasing opioid consumption. Evidence for the effect of all other included sociodemographic predictors on opioid use increase is inconclusive, given our relatively small sample size (N=346) and imprecise confidence intervals.

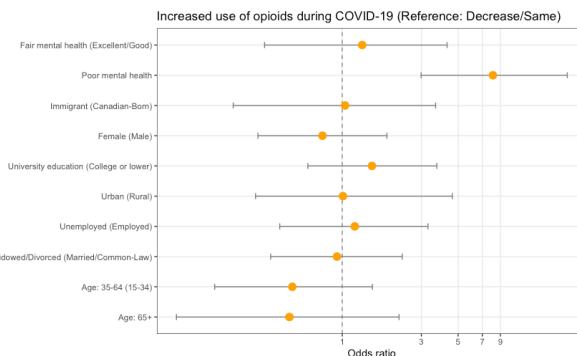


Figure 4. Forest plot of ORs for opioid use increase (compared to decrease/same level of use).

Odds of unmet need for care among opioid users. Our data shows that the odds of seeking help for substance misuse or mental health but being unable to find it follows similar patterns as the odds of increased opioid use. All else constant, unsuccessful help-seeking, and thereby unmet need for care among opioid users, is strongly associated with fair and poor mental health scores compared to opioid users in excellent or good mental health (Fair: 7.40, 95% CI 3.46–16.41; Poor: 19.48, 95% CI 8.21–48.83). The direction of causality for these associations, however, cannot be determined by this analysis. Further research is needed to explore potential reverse causality and to uncover the actual mechanism behind the relationship between mental health and inability to find satisfactory treatment for misuse. Moreover, single, widowed, or divorced individuals were about 80% more likely to be unsuccessful in seeking help for substance misuse or mental health, compared to their partnered counterparts, keeping all else constant (1.79, 95% CI 1.05–3.07). Our data also shows that older age appears to be a potential protective factor against unmet need. Based on Figure 3, it appears that females, university-educated individuals, and urban dwellers are more likely to be unsuccessful in finding help for substance misuse, compared to males, individuals with college education or lower, and rural dwellers, respectively. No significant difference in the odds of unmet need was observed by immigration status or employment.

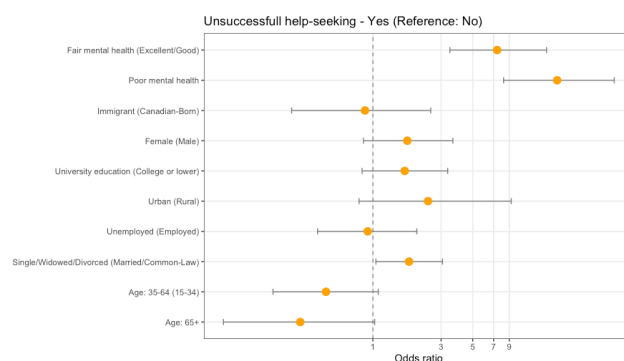


Figure 5. Forest plot of ORs for unsuccessful help-seeking for substance misuse and mental health among opioid users (compared to no unsuccessful help-seeking).

NPD use increase during COVID-19 and unmet treatment needs for substance misuse and mental health among NPD users

Figures 4 and 5 present the associations between NPD use change, ability to access treatment among NPD users and eight predictor variables (binary regression results tables are available in Appendix B).

Odds of NPD use increase. Much like with opioid use, we observe that increased use of NPDs during the COVID-19 pandemic is strongly associated with poor mental health. Keeping other covariates constant, compared to Canadians who self-rate their mental health as excellent or good, those with 'fair' and 'poor' mental health respectively are expected to be 3.3 and 5.7 times more likely to have increased their NPD use (Fair: 3.31, 95% CI 1.40–7.90; Poor: 5.73, 95% CI 2.20–14.97). Other predictor variables do not appear to affect the odds of increased NPD use in a statistically significant way, but we nonetheless see age as a potential protective factor. Compared to individuals aged 15 to 34, those over 65 have 70% lower odds of increasing their NPD use, with higher confidence intervals around the null effect (0.31, 95% CI 0.08–1.08). We observed no significant change in drug use patterns by immigration status, sex, education level, place of residence, employment, or marital status.

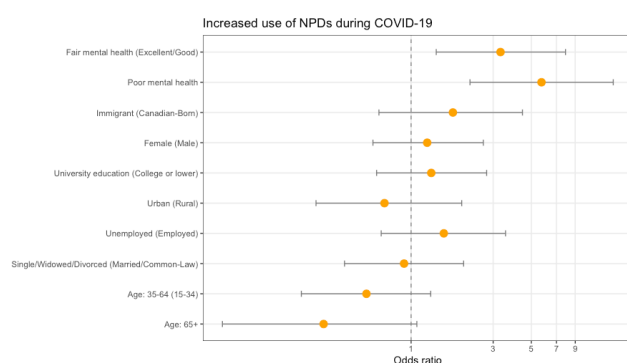


Figure 6. Forest plot of ORs for NPD use increase (compared to decrease/same level of use).

Odds of unmet need for care among NPD users. Our data shows that the odds of seeking help for substance misuse or mental health but being unable to find it follows similar patterns as the odds of increased NPD use. All else constant, unmet need for care among NPD users is strongly associated with fair and poor mental health scores compared to NPD users in excellent or good mental health (Fair: 3.85, 95% CI 1.83–8.19; Poor: 14.81, 95% CI 6.36–36.31). Additionally, single, widowed, or divorced individuals were about twice as likely to be unsuccessful in seeking help for substance misuse or mental health, compared to their married counterparts, keeping all else constant (2.06, 95% CI 1.19–3.61). Our data also shows that unemployed individuals are 54% more likely to be able to find help for substance misuse or mental health than employed individuals, all else constant (0.54, 95% CI 0.29–0.99). No significant difference in the odds of seeking help but being unable to find it was observed by immigration status, sex, education level, place of residence, or age.

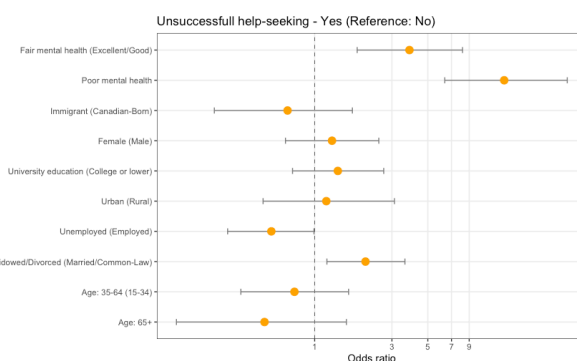


Figure 7. Forest plot of ORs for unsuccessful help-seeking for substance misuse and mental health among NPD users (compared to no unsuccessful help-seeking).

Differences between survey respondents who reported past drug use and those who did not

Figures 6 and 7 below present the results from our binary logit regressions estimating the odds of indicating past opioid and/or NPD drug use (i.e., stating that drug use increased, decreased, or stayed the same) compared to not responding (i.e., validly skipping the questions following a previous response of no recorded use) across our eight predictor variables and unmet need for care (see Appendix B for results tables). This analysis was conducted within our supplemental analytic samples of 3,932 and 3,934 complete cases with respect to our variables of interest, counting valid skip of our outcome variables as a valid response. The results from this analysis could be explained either as differences between respondents and non-respondents, or, when interpreted with caution, proxy results for predictors of drug use compared to no drug use, rather than predictors of response/non-response.

Overall, we observe that individuals with poor mental health scores are more likely to respond to questions related to their drug use, than individuals who self-rate their mental health as excellent or good. The odds of response to drug use questions are over 2.5 times higher for individuals with 'poor' compared to 'excellent' mental health, keeping other variables constant, for opioids (2.76, 95% CI 1.78–4.19) and NPDs (2.53, 95% CI 1.61–3.91), respectively. From Figures 6 and 7, we also see a gradient when it comes to different self-rated mental health scores and both drug use outcomes, whereby as mental health improves, the odds of response to drug questions gradually decrease.

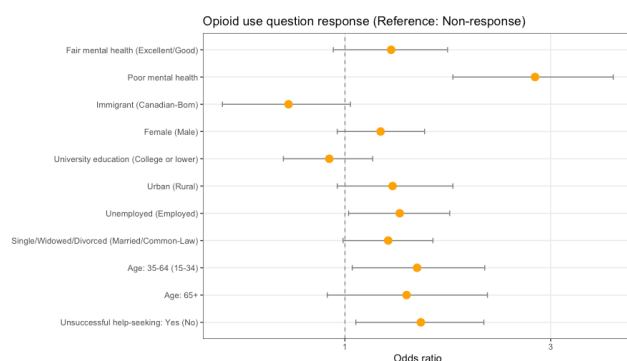


Figure 8. Forest plot of ORs for opioid use survey response (compared to valid skip).

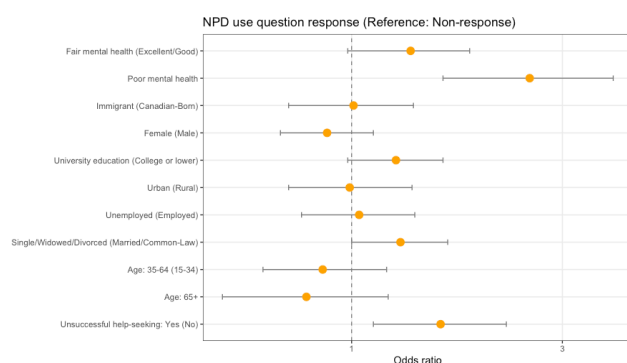


Figure 9. Forest plot of ORs for NPD use survey response (compared to valid skip).

The consistency of these results, especially given the body of existing research evidence of a negative relationship between mental health and substance use, points to the strong possibility that in this case, non-response is informative, and could be used as a proxy for drug use/non-use. Therefore, our results from proxy analysis suggest that individuals with poor mental health scores are more likely to have a history of drug use than those who self-report excellent mental health.

According to our data, single, divorced, or widowed individuals are almost 30% more likely to report opioid and NPD use compared to those who are married or cohabiting with partners, keeping other variables constant (opioids: 1.26, 95% CI 0.99–1.60; NPDs: 1.29, 95% CI 1.00–1.65). Our data also revealed that the odds of responding to opioid use questions were 34% higher for unemployed individuals compared to those who were employed during the COVID-19 pandemic at the time of interview, all else constant (1.34, 95% CI 1.02–1.75). Moreover, the odds of responding to both opioid and NPD use questions were over 50% higher for individuals who stated that they were unable to find needed help for substance use and mental health problems, all else kept constant (Opioids: 1.50, 95% CI 1.06–2.10; NPD: 1.59, 95% CI 1.12–2.24).

Gender, education level, and place of residence do not appear to affect response/nonresponse to both drug use questions in a statistically significant way according to the conventional boundaries of statistical significance at $p < 0.05$. However, based on the forest plots (Figures 6 and 7), we observe some trends that may be clinically important, albeit statisti-

cally insignificant. Accordingly, the odds of response to the NPD use change question are 26% higher for university-educated individuals compared to individuals with a college degree or lower (1.26, 95% CI 0.98–1.61). Similarly, landed immigrants seem to be about 26% less likely to report opioid use than native-born Canadians (0.74, 95% CI 0.52–1.03). We also observe opposing trends in the odds of opioid and NPD use response across age groups. While for opioid use, the odds of response appear to increase with age, for NPD use, the odds tend to decrease.

4 Discussion

Much of the research published during the initial stages of the COVID-19 pandemic focused on epidemiological modelling, hospitalizations and infection spread directly related to the SARS-COV-2 virus. However, as many indications are now pointing to the fact that we will likely have to learn to live with this virus for at least some years to come, attention must now shift to the consequences of living under the stress and uncertainties of lockdown measures and drastic changes to our lifestyles, work, and public policies.

We are only just starting to uncover the complex impact of the COVID-19 pandemic on mental health, health behaviors and social capital and how these outcomes differ across different sociodemographic characteristics. Individuals with substance use disorder (SUD) prior to the pandemic may have been particularly affected by the stress and hardship associated with the sudden lifestyle changes, and increased their drug use as a coping mechanism. Moreover, individuals affected by SUDs were already significantly more likely than the general population to suffer from a myriad of dangerous comorbidities, such as mental health problems, obesity, diabetes, HIV/AIDS as well as unhealthy eating and drinking habits (National Institute on Drug Abuse, 2018). Such individuals are also more likely to experience barriers to treatment and access to other integrated resources, while being at higher risk of COVID-19 transmission and worse disease progression.

This paper offers a first glance at opioid and non-prescription drug use pattern changes during the COVID-19 pandemic in Canada. We explored (a) the extent to which opioid and NPD use changed during the pandemic, and (b) whether individuals experienced barriers to access of substance misuse or mental health services. Mapping the changes in drug use, an avoidable risk factor in Canada, as well as the extent of unmet need for mental health and substance abuse services during the pandemic is crucial for effective and evidence-based design of future policy reforms.

Our findings indicate a strong relationship between poor mental health and co-occurring increasing drug use and inability to find help for substance misuse or mental health. These results not only confirm existing research documenting strong evidence of mental health and substance use comorbidity risk as well as stigma around addiction and unmet needs for treatment in Canada (Urbanoski et al., 2017), but also complement emerging studies and reports about negative changes in health behaviors and mental health of Canadians during COVID-19 (Zajacova et al. 2020, Gilmour, 2020, Rotermann, 2020). The

results highlight the need to improve access to integrated services for those who suffer from comorbid substance use and mental health problems. It is difficult, however, to establish the causal pathway of our findings (i.e., whether adverse mental health led to increased substance use, whether increased substance use led to poor mental health, or whether a common third factor, such as trauma, contributed to both).

Another crucial finding was that single, divorced, or widowed individuals were more likely to face unmet needs for care than their married or cohabiting counterparts. They were also significantly more likely to respond to both opioid and NPD consumption change questions, rather than skip the question following a response of 'no drug use' in a previous question. When interpreting this predictor of response as a predictor of drug use, the results are in line with recent research in the field, pointing to increased risk of drug use and loneliness among young and single individuals during the COVID-19 pandemic (Horigian et al., 2021). Single individuals may have been unable to date and socialize with friends, leading to loneliness, mental health problems and potentially drug use as a coping mechanism. This highlights the vast potential importance of social contact, intimacy, and physical company, all of which single individuals were deprived of during confinement. The relationship between loneliness, mental health and substance use has already been documented in research prior to the pandemic, but the risks of drug use and abuse associated with these factors are even more pressing now (Horigian et al., 2021).

We found no statistical evidence of a sex effect in drug use change, but females do appear to report more difficulties accessing services for substance misuse, despite seeking help. This points to the need to design substance use interventions with gender in mind and sensitive to a potentially differential level of stigma associated with drug use for females, compared to males, which may lead to differential help-seeking behavior by gender.

While older age appears to be a protective factor against an increase in consumption of both opioids and NPDs during the COVID-19 pandemic, as well as unmet needs in both populations, we observed opposing trends in the odds of opioid and NPD use question response across age groups. This suggests that two separate opioid and NPD use trends may be at play. While older individuals may be at higher risk for misuse of opioids, as a potential coping mechanism with increasing discomfort and chronic pain associated with aging, NPD use may be more concentrated among younger age groups, to cope with life-related stressors, such as education, financing, and work-family balance.

The findings presented from our analysis are limited by the cross-sectional design of the CPSS data series. We are only able to present a time-specific snapshot, rather than uncover longitudinal trends over the period of the pandemic and prior to it. Furthermore, due to space and scope limitations, we chose to limit our analysis to an exploration of the risk factors associated with increased drug consumption during the COVID-19 pandemic. Investigation of potential protective factors associated with decreased drug use is also substantially

important to the community of researchers and decision makers and should be addressed in future studies.

An additional potential limitation of our analysis is that all our outcomes of interest were subject to potential recall bias and incorrect assessment of individual changes in drug use at the time of the interview. Moreover, we cannot assess the magnitude of the self-reported, subjective changes. Similarly, without follow-up, we are unable to establish the proportion of individuals who have previously not consumed any drugs and initiated consumption only during the COVID-19 pandemic. This makes it difficult to uncover the precise mechanisms through which the changes in drug use occurred, and individual characteristics that contribute to increased risk of initiating drug consumption. Likewise, we were unable to uncover if, and how, mental health outcomes were affected as the pandemic progressed. There are also notable limitations related to the risk of bias. Although reporting bias and social desirability bias appear to be a serious possibility for all ages due to the sensitive nature of the subject, the survey respondents are assured that their responses will remain secure, confidential, and anonymous. Moreover, due to a relatively high non-response rate to CPSS6 survey among CPSS panelists (46.6%), it is likely that those who volunteered to respond differ in meaningful ways from those who did not, which may result in selection bias.

Another issue pertains to the wording of the opioid question, which includes both prescribed and non-prescribed opioids, precluding a differentiated analysis of legal and illicit consumption change patterns, which may well have differed during the pandemic. Due to the phrasing of the survey question, however, it is also possible that many respondents incorrectly stated that they have never used opioids if they were unaware that this category also included prescribed and non-prescribed opioids such as OxyContin, Codeine, or Tylenol. While the questionnaire provided an explanatory list of the description of included drugs, some respondents may have skipped the explanation, unsuspecting that some opioids are legal and commonly prescribed for severe pain management. It was very interesting in this context to find overwhelmingly similar associations between our sociodemographic determinants and both outcomes, (licit or illicit) opioid use change and illicit NPD use change.

Further, data availability posed additional challenges for our analysis and the robustness of results. For example, variables related to ethnicity, indigenous status, homelessness, and province of residence were not available as part of the CPSS 6 data series. Moreover, the reported variable for sex of respondents is dichotomized into 'male' and 'female' only, leaving individuals who do not identify with either of these categories invisible. These covariates would have been a valuable addition to our regression analyses, for they would allow us to control for a well-documented differential impact of COVID-19 among different ethnic groups (Rentsch et al., 2020), vulnerable groups, such as homeless individuals (Perri et al., 2020) or gender/sexual minorities (Parent et al., 2018), and differential burden of the pandemic across different Canadian provinces (Detsky & Bogoch, 2020). We appreciate the difficulties in designing surveys such as CPSS, but would welcome if future

waves collected more equity, diversity and inclusion variables and presented them as part of their PUMF datasets.

5 Conclusion

This paper explored the extent to which COVID-19 affected opioid and NPD use among Canadians and examined their help-seeking behavior for substance misuse or mental health, controlling for a range of sociodemographic determinants. We document significant associations between perceived mental health and increased drug use, as well as an inability to access substance misuse or mental health resources. This highlights the importance of providing enhanced integrated services for mental health and substance abuse comorbidities in Canada. As we slowly emerge from the pandemic, we should start considering how to provide needed support and tailored interventions to address additional consequences of the COVID-19 pandemic, the stress, isolation, uncertainty, and changes to our lifestyles, such as changes to drug use, among other health behaviors.

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Appendix A: Relevant Parts of the Questionnaire

Substance use — opioids (OPI)

OPI_BEG

External Variables required:

O The following questions are about your opioid use.

PI In the past 30 days, have you used any prescribed or non-prescribed drugs containing opioids?

-
Q
0
5

ON-SCREEN HELP: By opioids, we mean products such as codeine, oxycodone, OxyContin, OxyNeo, hydromorphone, Dilaudid, morphine, fentanyl, methadone, Tylenol 1, 2, 3 or 4, 292s, Robaxacet-8, Percocet, Percodan, Demerol, AC&C, Mersyndol, Calmylin, and heroin for either medical or non-medical purposes.

1 Yes

2 No, not during the past 30 days

3 Never used opioid products

(Don't know, Refusal not allowed)

OPI_C10

If OPI_Q05 = 3, go to OPI_END.

Otherwise, go to OPI_Q10.

OPI_Q
10On average, over the course of the COVID-19 pandemic, how has your use of opioids changed when comparing to **before** the pandemic?

1 Increased

2 Decreased

3 Not applicable

4 Stayed about the same

(Don't know, Refusal not allowed)

Substance use — non-prescribed drugs (NPD)

NPD_BEG

External Variables required:

N The following questions are about your non-prescribed drug use.

P In the past 30 days, did you use any non-prescribed drugs?

D

-
Q
0
5ON-SCREEN HELP: **Exclude** cannabis and opioids.

For the purpose of this survey, non-prescribed drugs refer to the use of various substances, such as cocaine, crystal meth, amphetamines, ecstasy, hallucinogens, as well as the misuse of over-the-counter drugs.

By "misuse of over-the-counter drugs", we mean **for example**, for the purpose of getting high, use of laxatives for weight loss, taking higher than recommended doses, **etc.**

1 Yes

2 No, not during the past 30 days

3 Never used any non-prescribed drugs

(Don't know, Refusal not allowed)

NPD_C10

If NPD_Q05 = 3, go to NPD_END.

Otherwise, go to NPD_Q10.

NPD_
Q10On average, over the course of the COVID-19 pandemic, how has your use of non-prescribed drugs changed when comparing to **before** the pandemic?

1 Increased

2 Decreased

3 Not applicable

4 Stayed about the same

(Don't know, Refusal not allowed)

HS_ Since the beginning of the COVID-19 pandemic, was there ever a time when you felt that you needed help for your emotions, mental health or use of alcohol or drugs, but you didn't receive it?

Q20

ON-SCREEN HELP: **Include** alcohol and prescribed and non-prescribed drugs.

1 Yes

2 No

(Don't know, Refusal not allowed)

HS_END

Appendix B: Binomial Regression Analyses

Binomial logistic regressions of opioid use changes and ability of opioid users to access help (Observations: 346)

Independent variables	vari-	OR (Opioid use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Mental Health	Excellent/Good	(reference)	(reference)
	Fair	1.32 (0.34–4.29, p=0.660)	7.40 (3.46–16.41, p<0.001)
	Poor	8.11 (2.99–22.75, p<0.001)	19.48 (8.21–48.83, p<0.001)
Immigration status	Born in Canada	(reference)	(reference)
	Landed immigrant	1.04 (0.22–3.66, p=0.952)	0.88 (0.27–2.54, p=0.825)
Sex	Male	(reference)	(reference)
	Female	0.76 (0.31–1.86, p=0.541)	1.74 (0.86–3.63, p=0.127)
Education level	College or lower	(reference)	(reference)
	University education	1.51 (0.62–3.72, p=0.364)	1.67 (0.84–3.34, p=0.144)
Place of residence	Rural	(reference)	(reference)
	Urban	1.01 (0.30–4.61, p=0.987)	2.43 (0.80–9.29, p=0.147)
Employment status	Employed	(reference)	(reference)
	Unemployed	1.19 (0.42–3.29, p=0.733)	0.92 (0.41–2.03, p=0.846)
Marital status	Married/Common Law	(reference)	(reference)
	Single/Widowed/Divorced	0.93 (0.37–2.30, p=0.882)	1.79 (1.05–3.07, p=0.033)
Age group	15–34	(reference)	(reference)
	35–64	0.50 (0.17–1.52, p=0.202)	0.47 (0.20–1.09, p=0.075)
	65+	0.48 (0.10–2.20, p=0.348)	0.31 (0.09–1.03, p=0.059)

Binomial logistic regressions of NPD use changes and ability of NPD users to access help (Observations: 308)

Independent variables	vari-	OR (NPD use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Mental Health	Excellent/Good	(reference)	(reference)
	Fair	3.31 (1.40–7.90, p=0.006)	3.85 (1.83–8.19, p<0.001)
	Poor	5.73 (2.20–14.97, p<0.001)	14.81 (6.36–36.31, p<0.001)
Immigration status	Born in Canada	(reference)	(reference)
	Landed immigrant	1.75 (0.65–4.44, p=0.247)	0.68 (0.24–1.71, p=0.430)
Sex	Male	(reference)	(reference)
	Female	1.24 (0.60–2.63, p=0.565)	1.28 (0.66–2.49, p=0.469)
Education level	College or lower	(reference)	(reference)
	University education	1.31 (0.63–2.75, p=0.471)	1.39 (0.73–2.67, p=0.321)
Place of residence	Rural	(reference)	(reference)
	Urban	0.70 (0.28–1.97, p=0.478)	1.18 (0.48–3.11, p=0.724)
Employment status	Employed	(reference)	(reference)
	Unemployed	1.55 (0.67–3.54, p=0.300)	0.54 (0.29–0.99, p=0.051)
Marital status	Married/Common Law	(reference)	(reference)
	Single/Widowed/Divorced	0.91 (0.41–2.02, p=0.823)	2.06 (1.19–3.61, p=0.011)
Age group	15–34	(reference)	(reference)
	35–64	0.55 (0.23–1.30, p=0.166)	0.75 (0.35–1.62, p=0.455)
	65+	0.31 (0.08–1.08, p=0.075)	0.49 (0.14–1.57, p=0.239)

Binomial logistic regressions of opioid and NPD question response (Observations: 3,932 / 3,934)

Independent variables	vari-	OR (Opioid use response), Ref: valid skip/N.A.	OR (NPD use response), Ref: valid skip/N.A.
Mental Health	Excellent/Good	(reference)	(reference)
	Fair	1.28 (0.94–1.73, p=0.116)	1.36 (0.98–1.85, p=0.060)
	Poor	2.76 (1.78–4.19, p<0.001)	2.53 (1.61–3.91, p<0.001)
Immigration status	Born in Canada	(reference)	(reference)
	Landed immigrant	0.74 (0.52–1.03, p=0.079)	1.01 (0.72–1.38, p=0.965)
Sex	Male	(reference)	(reference)
	Female	1.21 (0.96–1.53, p=0.114)	0.88 (0.69–1.12, p=0.290)
Education level	College or lower	(reference)	(reference)
	University education	0.92 (0.72–1.16, p=0.468)	1.26 (0.98–1.61, p=0.070)
Place of residence	Rural	(reference)	(reference)
	Urban	1.29 (0.96–1.78, p=0.102)	0.99 (0.72–1.37, p=0.948)
Employment status	Employed	(reference)	(reference)
	Unemployed	1.34 (1.02–1.75, p=0.035)	1.04 (0.77–1.39, p=0.800)
Marital status	Married/Common Law	(reference)	(reference)

Independent variables	vari-	OR (Opioid use response), Ref: valid skip/N.A.	OR (NPD use response), Ref: valid skip/N.A.
Age group	Single/Widowed/Divorced	1.26 (0.99–1.60, p=0.055)	1.29 (1.00–1.65, p=0.049)
	15–34	(reference)	(reference)
	35–64	1.47 (1.04–2.11, p=0.033)	0.86 (0.63–1.20, p=0.363)
Unsuccessful seeking	65+	1.39 (0.91–2.14, p=0.136)	0.79 (0.51–1.21, p=0.269)
	No	(reference)	(reference)
	Yes	1.50 (1.06–2.10, p=0.021)	1.59 (1.12–2.24, p=0.009)

Appendix C: Supplemental Binomial Logistic Models (Subgroup Analyses)

Demographic-only model, opioid outcomes (Observations: 346)

Demographic variables		OR (Opioid use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Immigration status	Born in Canada	(reference)	(reference)
	Landed immigrant	1.06 (0.43–2.39, p=0.886)	0.68 (0.24–1.65, p=0.424)
Sex	Male	(reference)	(reference)
	Female	0.67 (0.38–1.19, p=0.173)	1.49 (0.84–2.70, p=0.174)
Place of residence	Rural	(reference)	(reference)
	Urban	1.32 (0.61–3.18, p=0.506)	2.13 (0.94–5.49, p=0.088)
Age group	15–34	(reference)	(reference)
	35–64	0.70 (0.33–1.56, p=0.362)	0.33 (0.17–0.65, p=0.001)
	65+	0.69 (0.28–1.70, p=0.408)	0.16 (0.06–0.39, p<0.001)

Socio-economic-only model, opioid outcomes (Observations: 346)

Socio-economic variables	vari-	OR (Opioid use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Mental health	Excellent/Good	(reference)	(reference)
	Fair	1.40 (0.37–4.43, p=0.590)	8.32 (3.97–18.06, p<0.001)
	Poor	9.07 (3.56–23.74, p<0.001)	22.86 (10.01–55.22, p<0.001)
Education level	College or lower	(reference)	(reference)
	University education	1.37 (0.57–3.28, p=0.474)	1.73 (0.90–3.39, p=0.103)
Employment status	Employed	(reference)	(reference)
	Unemployed	1.02 (0.42–2.44, p=0.963)	0.72 (0.37–1.38, p=0.323)
Marital status	Married/Common Law	(reference)	(reference)
	Single/Widowed/Divorced	0.98 (0.40–2.31, p=0.958)	1.15 (0.60–2.21, p=0.672)

Demographic-only model, NPD outcomes (Observations: 308)

Demographic variables		OR (NPD use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Immigration status	Born in Canada	(reference)	(reference)
	Landed immigrant	1.84 (0.87–3.79, p=0.104)	0.58 (0.22–1.33, p=0.226)
Sex	Male	(reference)	(reference)
	Female	0.83 (0.47–1.48, p=0.530)	1.35 (0.77–2.40, p=0.295)
Place of residence	Rural	(reference)	(reference)
	Urban	1.09 (0.51–2.49, p=0.829)	1.20 (0.58–2.65, p=0.632)
Age group	15–34	(reference)	(reference)
	35–64	0.34 (0.18–0.64, p=0.001)	0.60 (0.32–1.13, p=0.108)
	65+	0.19 (0.07–0.46, p<0.001)	0.20 (0.07–0.50, p=0.001)

Socio-economic-only model, NPD outcomes (Observations: 308)

Socio-economic variables	vari-	OR (NPD use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Mental health	Excellent/Good	(reference)	(reference)
	Fair	3.97 (1.72–9.27, p=0.001)	4.32 (2.09–9.03, p<0.001)

Socio-economic variables	vari-	OR (NPD use increase), Ref: decrease/same	OR (unsuccessful help-seeking – yes), Ref: no
Education level	Poor	6.37 (2.51–16.19, p<0.001)	16.23 (7.08–39.14, p<0.001)
	College or lower	(reference)	(reference)
Employment status	University education	1.29 (0.63–2.68, p=0.491)	1.35 (0.71–2.58, p=0.362)
	Employed	(reference)	(reference)
Marital status	Unemployed	1.29 (0.61–2.70, p=0.503)	0.64 (0.31–1.26, p=0.206)
	Married/Common Law	(reference)	(reference)
	Single/Widowed/Divorced	0.89 (0.43–1.84, p=0.760)	1.37 (0.72–2.60, p=0.342)

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